Experiment No. 9

Title

Understanding Different Types of Shells and Writing A Shell Script

Aim

The problem is to identify different types of shells available in Linux, understand their characteristics, and write and execute a simple shell script using Bash, including the use of variables.

Theory

Shell In Linux

A shell is a command-line interpreter that processes user commands and communicates with the kernel. It allows users to execute commands, manage files, and write scripts for automation.

Types Of Shells

Shell	Description
sh	Original Bourne Shell
bash	Bourne Again Shell (default)
ksh	Korn Shell
csh	C Shell
zsh	Z Shell

Bash Shell

Bash is an enhanced version of sh that supports:

- Command history
- Job control
- Scripting with variables and control structures

Shell Scripts

A shell script is a file containing a sequence of commands executed by the shell. It typically starts with a shebang (#!/bin/bash) that specifies the interpreter.

COMMANDS USED-

`cat /etc/shells`	List available shells
`echo $SHELL`	Display current shell
`bash`	Start Bash shell
`sh`	Start Bourne shell
`chmod`	Change file permissions
`./script.sh`	Execute shell script

Procedure / Steps

A. Identifying Different Shells

Open the terminal.

List available shells:

cat /etc/shells

Check the current shell:

echo $SHELL

B. Switching Between Shells

Switch to Bourne shell:

sh

Return to Bash:

Exit

```
┌──(setika㉿kali)-[~]
└─$ cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/usr/bin/dash
/usr/bin/pwsh
/opt/microsoft/powershell/7/pwsh
/usr/bin/screen
/usr/bin/tmux
/bin/zsh
/usr/bin/zsh

┌──(setika㉿kali)-[~]
└─$ echo $SHELL
/usr/bin/zsh

┌──(setika㉿kali)-[~]
└─$ sh
$ exit

┌──(setika㉿kali)-[~]
└─$ nano simple.sh
```

C. Writing a Simple Bash Script

Create a new script file:

nano simple.sh

Enter the following script:

#!/bin/bash

echo "Welcome to Linux Shell Scripting"

name="Student"

echo "Hello $name" Save and exit the editor.

```
Session   Actions   Edit   View   Help
  GNU nano 8.7                                    simple.sh
#!bin/bash
echo "welcome to linux shell scripting"
name="student"
echo "hello $name"
```

D. Executing the Script

Provide execute permission:

chmod +x simple.sh

Run the script:

./simple.sh

```
┌──(setika㉿kali)-[~]
└─$ nano simple.sh

┌──(setika㉿kali)-[~]
└─$ chmod +x simple.sh

┌──(setika㉿kali)-[~]
└─$ ./simple.sh
```

Outcomes

- ✓ List of available shells displayed.
- ✓ Current shell identified.
- ✓ Simple Bash script executed successfully.
- ✓ Output displays variable values correctly.

Result

- ❖ Different types of Linux shells were identified and a simple Bash shell script using variables was successfully written and executed.